EXAM ANSWER BOOKLET - QUICK WRITING GUIDE

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Project: Student Study & Task Hub | **Live URL:** https://nishulucyna.github.io/student_study_hub/

1. APPLICATION NAME & WHAT IT DOES

Name: Student Study & Task Hub

What it does: A clean Flutter web application that helps students track study tasks. Students can view tasks, mark them complete, add new tasks through a validated input form, toggle Light/Dark themes, and load motivational study quotes from a REST API.

2. PURPOSE & OBJECTIVES

To provide a simple, distraction-free study organizer while covering all syllabus lab topics:

- Basic UI widgets and layouts (Row, Column, Stack)
- Responsive design adapting between mobile and desktop screens
- Multi-screen navigation using named routes
- State management using `setState` and `Provider`
- Form validation and error handling
- UI animations (fade, slide, and animated container)
- Fetching data asynchronously from a REST API using `http`
- Unit and widget testing for verification

3. TOOLS & TECHNOLOGIES USED

- **Flutter SDK & Dart:** For developing the cross-platform UI and logic.
- **Provider Package:** For app-wide state management (task list and theme).
- **HTTP Package:** To fetch data asynchronously from a REST API.
- **Flutter Test:** For automated unit and widget testing.
- **GitHub Actions & GitHub Pages:** For automated CI/CD build and free web hosting.

4. MAJOR FEATURES IMPLEMENTED

- **Responsive UI:** Uses `MediaQuery`. On wide screens (>700px), tasks show in a 2-column Grid; on mobile, in a 1-column List.
- **Multi-Screen Navigation:** Configured named routes (`/` , `/add-task` , `/api-quotes`) using `Navigator.pushNamed` .
- **Form Validation:** Validates that title, subject, and email are not empty and email has valid format (`@` , `.`).
- **REST API Integration:** Fetches live quotes from `https://dummyjson.com/quotes?limit=6` using `FutureBuilder` .
- **Animations:** Smooth entry fade/slide transitions and an animated expanding banner.

5. HOW IT WAS DEVELOPED & DEPLOYED

- **Development:** Built with modular screens (Home, Add Task, Quotes) and a custom reusable task card widget. Tested with `flutter test` .
- **Deployment:** Pushed to GitHub. A GitHub Actions workflow runs tests, builds the web bundle (`flutter build web --base-href "/student_study_hub/" --release`), and deploys the `build/web` folder to GitHub Pages.

Tip for exam: Write these 5 short headings directly in your answer booklet. They are concise, clear, and cover every required grading point.